

PCI AI • FORMULA SHEET

Project Controls Formulas

The 16 that decide whether your forecast
is defensible

CPI • SPI • EAC • TCPI • TF • EMV

PROJECT CONTROLS INSTITUTE GLOBAL

First edition • 2026 • AI proposes. The professional disposes.

The formula is never the hard part

Every controls professional can write **$CPI = EV \div AC$** .

Far fewer can say what makes that number wrong, which is almost always the input rather than the expression.

This deck gives the formula, and the mistake it attracts.

The rule underneath all of it. A correct formula applied to a compromised input produces a confident, well-formatted, entirely wrong answer. Those are the expensive ones.

NOTATION

Two symbols. One page. Constant confusion.

PV means **Planned Value** in earned value.

PV means **present value** in discounting.

Both appear in the same cost report every month.

Write it out. Use "present value" in words, or **PV(x)**, whenever you are discounting. This pair collides more often than any other in the discipline.

Where you are

$$CV = EV - AC$$

$$SV = EV - PV$$

$$CPI = EV \div AC$$

$$SPI = EV \div PV$$

Negative variance is adverse. Index below 1.00 is adverse.

THE ACCRUAL RULE

The number that breaks all four

$$\text{Cost to date} = \text{Actuals} + \text{Accruals}$$

Every index above divides by **AC**. If **AC** is invoiced cost only, work has been performed that nobody has charged you for yet.

CPI overstates performance. **EAC** understates cost. Both errors are invisible in the report.

Before you compute anything: reconcile goods-received-not-invoiced against the accrual booked. A persistent gap is not a timing quirk. It is a wrong forecast.

WORKED · 420M PROGRAMME, MONTH 14

What one missed accrual does

Invoiced cost only → **CPI** 0.957 → **EAC** 438.9m

Accrual booked, +12.4m → **CPI** 0.887 → **EAC** 473.3m

The 34.4m swing exceeded the entire 21m contingency. The board had been told the project was uncomfortable but contained. It was neither — and the forecasting was never wrong. The input was.

Three answers. All correct.

$$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV})$$

$$\text{EAC} = \text{BAC} \div \text{CPI}$$

$$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV}) \div (\text{CPI} \times \text{SPI})$$

Same data. Three defensible numbers. The arithmetic is trivial; the assumption is the professional judgement.

DECISION AID

So which one

IF	THEN
The variance was a closed, one-off event	$AC + (BAC - EV)$
Performance to date will persist	$BAC \div CPI$
Cost and schedule pressure both persist	$AC + (BAC - EV) \div (CPI \times SPI)$
Remaining scope differs materially	Bottom-up ETC — indices do not apply

Publish the range, not the point. A single EAC quoted to the nearest thousand claims a precision the method cannot support.

TCPI

The reality check nobody runs

$$\text{TCPI} = (\text{BAC} - \text{EV}) \div (\text{BAC} - \text{AC})$$

This is the cost efficiency required on all remaining work to still land on budget.

Compare it to the **CPI** you have actually achieved.

If TCPI is far above CPI, the budget is asserting an improvement that nobody has yet demonstrated. On the worked case: TCPI 1.08 against a CPI of 0.89 — the plan needs a 22% efficiency gain on work running 11% below plan.

EARNED SCHEDULE

Why SV lies to you at the end

SV and **SPI** are denominated in **currency**.

At completion **EV = PV**, so **SV** goes to zero and **SPI** goes to 1.00 — however late you finished.

$$ES = M + (EV - PV_M) \div (PV_{M+1} - PV_M)$$

$$SV(t) = ES - AT$$

$$SPI(t) = ES \div AT$$

These do not converge. Late in a project, earned schedule is the honest measure and the currency-based one is not.

Float, two ways

$$EF = ES + \text{Duration}$$

$$LS = LF - \text{Duration}$$

$$TF = LS - ES = LF - EF$$

$$FF = \min(\text{successor ES}) - EF$$

A free integrity check. Total float computed from the starts must equal total float computed from the finishes. If the two disagree, your forward or backward pass is wrong.

ESTIMATING

Three points, one duration

$$tE = (O + 4M + P) \div 6$$

$$\sigma = (P - O) \div 6$$

PERT weights the most likely case four times. The triangular alternative, $(O + M + P) \div 3$, does not — and gives a different answer.

Watch for: three points supplied by one person. That is one estimate wearing three hats, and it produces a false sense of range.

PATH UNCERTAINTY

**Variance adds.
Deviation does not.**

$$\sigma^2 \text{ path} = \Sigma \sigma^2 \text{ along the path}$$

Sum the **variances** along the path, then take the square root at the end.

Summing standard deviations directly overstates path uncertainty, often substantially.

Where contingency should come from

EMV = Probability × Impact
Contingency $\approx \Sigma$ EMV, or P80 – P50
from simulation

A percentage of budget is not an analysis. If contingency does not reconcile to quantified risk exposure, it is a habit. And **EMV** describes a portfolio of risks — applied to one binary event, the expected value is a number that will never occur.

PROGRESS MEASUREMENT

Never accept this number

WORK PACKAGE	METHOD
Spans one period	0/100
Two periods, discrete ends	50/50
Countable homogeneous output	Units complete
Long, with verifiable stages	Milestone weighting
No discrete output	Level of effort — never the default

Fix the method before work starts. And never accept a percentage complete supplied by the party being measured. That figure is a commercial position, not a measurement.

MOST MISAPPLIED • 1-5

The ten that cost the most

1. **CPI** computed on invoiced cost, with no accruals
2. **SV** read as a schedule measure late in a project
3. A single **EAC** presented as the forecast
4. **TCPI** never compared against the **CPI** achieved
5. Percentage complete supplied by the measured party

MOST MISAPPLIED · 6-10

And the other five

1. Standard deviations summed along a schedule path
2. **EMV** applied to a single binary event
3. Contingency set as a percentage of budget
4. Crashing an activity that is not on the critical path
5. Management reserve included inside **BAC**

Every one of these produces a report that passes review. That is precisely why they are expensive.

The full sheet is free

This deck is 16 of them. The complete reference carries every formula in the discipline — accounting, budgeting, cost, earned value, scheduling, risk, commercial and working capital — each with the mistake it attracts, the Knowledge Area where it is worked in full, and reproducible worked examples.

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